

Name: _____
ID Number: _____

Math 166
U-Substitution Gateway

1. Evaluate the following integral, and clearly show all your work. Then, put your final answer in the box provided. You might lose credit if your answer is not clearly written in the provided box.

$$\int_2^e \frac{(\ln(x))^2}{x} dx$$

Final answer =

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Math 166
Algebra Gateway 12:30 pm

Directions: Answer the following algebra questions. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

1. Simplify into a single fraction: $\frac{3}{4} + \frac{1}{3}$

2. Simplify into an expression with a single x :

$$\frac{x^{-3}}{x^2}$$

3. Find integers a and b where:

$$\frac{1}{x^2 + 2x + 2} = \frac{1}{(x + a)^2 + b}$$

$a =$ _____, $b =$ _____

4. Find all values of x where:

$$x + \frac{1}{x} = 2$$

$x =$ _____

5. For which value of a is the following true?

$$\ln(x^a) + 2 \ln(x) = \ln(x)$$

$a =$ _____

For the remaining questions, **check every answer** that is equivalent to the original expression, **and leave blank** every answer that is not equivalent. You will only get credit if you check all equivalent answers.

6. Original: $\ln(x^2)$. Options:

☐ $2 \ln(x)$ ☐ $(\ln(x))^2$ ☐ $\ln(x) \cdot \ln(x)$ ☐ $\ln(x) + \ln(x)$

7. Original: $\ln(xy/z)$. Options:

☐ $\ln(x) + \ln(y) - \ln(z)$ ☐ $\ln(xy)/\ln(z)$ ☐ $-\ln(z/xy)$ ☐ $x \ln(y/z)$

8. Original: $\frac{x}{x+2}$ for $x \neq 0$ and $x \neq 2$. Options:

☐ $\frac{1}{1 + 2/x}$ ☐ $1 - \frac{2}{x-2}$ ☐ $\left(\frac{x+2}{x}\right)^{-1}$ ☐ $1 + \frac{x}{2}$

9. Original: $\frac{1}{\sqrt{7}}$. Options:

☐ $\frac{7}{\sqrt{7}}$ ☐ $\frac{\sqrt{7}}{7}$ ☐ $7^{-1/2}$ ☐ $7^{1/2}$

10. Original: $\sqrt{100 - 64}$. Options:

☐ 6 ☐ $\sqrt{100} - \sqrt{64}$ ☐ 2 ☐ $\sqrt{36}$

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Math 166
Derivative Gateway 12:30 pm

Directions: Compute the following derivatives. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

1. $\frac{d}{dx}(x^2 - e)$

2. $\frac{d}{dx} \tan x$

3. $\frac{d}{dx} 2x^{1/2}$

4. $\frac{d^2}{dt^2} \sin(4t)$

5. $\frac{d}{dx} \sec x$

6. $\frac{d}{du} u \cos u$

7. $\frac{d}{dx} e^{\sin x}$

8. $\frac{d}{dx} (8x - 19)^{40}$

9. $f'(x)$ if $f(x) = \sin(\cos(x))$

10. $\frac{d}{dx} (f(x) \cdot g(x))$ in terms of $f(x)$ and $g(x)$
